STAT201 · STATISTICS · LECTURE 11

Hypothesis Testing II: Comparing Groups, χ^2 and ANOVA

Comparing two means and proportions, paired data, contingency tables and comparing three or more groups

Syllabus topic 11 · week 11 · 4 academic hours

LECTURE GOALS

- choose between independent and paired samples and carry out the corresponding t-test;
- compare two proportions with a z-test and construct a CI for the difference;
- test the independence of two categorical attributes and goodness of fit to a distribution using the χ^2 test;
- carry out a one-way analysis of variance (ANOVA) and interpret the F-statistic;
- judge the practical significance of differences and avoid typical mistakes (multiple comparisons, the wrong choice of test).

1 How to choose a test

The logic of hypothesis testing (Lecture 10) remains the same; the statistic and its distribution change. The choice depends on the type of data and the plan of collection:

Question	Test
Two means, independent groups	two-sample t-test (Welch)
Two measurements on the same objects (before/after)	paired t-test on the differences
Two proportions	two-sample z-test for proportions
Relationship between two categorical attributes	χ^2 test of independence
Does a distribution match a given one	χ^2 goodness-of-fit test
Three or more means	one-way ANOVA (F-test)

2 Two means: independent samples

Let $\bar{x}_1, s_1, n_1$ and $\bar{x}_2, s_2, n_2$ be obtained from two independent populations. $H_0 : \mu_1 - \mu_2 = 0$. The standard error of the difference (variances of independent variables add):

$$\mathrm{SE}(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \qquad t = \frac{\bar{x}_1 - \bar{x}_2}{\mathrm{SE}}, \tag{2.1}$$

and the degrees of freedom (the Welch–Satterthwaite formula)

$$\nu \approx \frac{\left(s_1^2/n_1 + s_2^2/n_2\right)^2}{\dfrac{(s_1^2/n_1)^2}{n_1 - 1} + \dfrac{(s_2^2/n_2)^2}{n_2 - 1}}. \tag{2.2}$$

This is Welch's test: it does not require equal variances and is the default setting in modern software. (If the variances are assumed equal, one can use the pooled estimate $s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2}$ and $\nu = n_1 + n_2 - 2$.) The CI for the difference: $(\bar{x}_1 - \bar{x}_2) \pm t_{\alpha/2,\nu}\,\mathrm{SE}$.

EXAMPLE 1. TRANSACTION TIME AT TWO BRANCHES

Branch 1: $n_1 = 12$, $\bar{x}_1 = 14.2$ min, $s_1 = 3.1$. Branch 2: $n_2 = 15$, $\bar{x}_2 = 12.1$, $s_2 = 2.6$. $H_0 : \mu_1 = \mu_2$, $H_1 : \mu_1 \neq \mu_2$, $\alpha = 0.05$. $\mathrm{SE} = \sqrt{3.1^2/12 + 2.6^2/15} = \sqrt{0.801 + 0.451} = 1.119$; $t = \dfrac{2.1}{1.119} = 1.88$; $\nu = 21.5$. $p = 0.074 > 0.05$ — at the 5% level the difference is not proven. The 95% CI of the difference: $2.1 \pm 2.077 \cdot 1.119 = (-0.22;\ 4.42)$ — contains 0, consistent with the test. With a pooled variance we would get almost the same ($t = 1.92$, $p = 0.067$). A one-sided test ($H_1 : \mu_1 > \mu_2$) would give $p = 0.037$, but, as said, the direction is chosen before the data.

3 Paired data

If the measurements relate to the same objects (before and after training; two methods on the same customers), the samples are dependent. Then we compute the *differences* $d_i = x_i^{\text{after}} - x_i^{\text{before}}$ and apply a one-sample t-test to them:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad \nu = n - 1. \tag{3.1}$$

EXAMPLE 2. STAFF TRAINING

Scores of 8 employees before training: $62, 70, 55, 68, 74, 60, 66, 71$; after: $66, 73, 58, 69, 79, 62, 70, 72$. Differences d: $4, 3, 3, 1, 5, 2, 4, 1$; $\bar{d} = 2.875$, $s_d = 1.458$, $\mathrm{SE} = 0.515$. $t = 5.58$, $\nu = 7$, $p = 0.0008$. We reject $H_0 : \mu_d = 0$: training raised the score by 2.9 on average (95% CI: $(1.66;\ 4.09)$).

COMMON MISTAKE

Paired data must not be analysed as independent. For the same data Welch's test gives $t = 0.89$, $p = 0.39$ — the effect "gets lost" because of the large spread between employees (which the paired design eliminates). The converse also holds: independent samples must not be "paired" arbitrarily.

4 Two proportions

$H_0 : p_1 = p_2$. Under H_0 both proportions estimate the same p, so the test uses the pooled proportion $\hat{p} = \frac{x_1+x_2}{n_1+n_2}$:

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}. \tag{4.1}$$

For a CI of the difference the separate error $\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$ is used.

EXAMPLE 3. TWO MAILING VARIANTS

Offer A: 120 responses out of 400 ($\hat{p}_1 = 0.300$); offer B: 150 out of 420 ($\hat{p}_2 = 0.357$). $\hat{p} = 270/820 = 0.329$, SE $= \sqrt{0.329 \cdot 0.671 \cdot (1/400 + 1/420)} = 0.0328$. $z = \dfrac{-0.0571}{0.0328} = -1.74$, $p = 0.082$ (two-sided). The difference is not proven at the 5% level. 95% CI of the difference $p_1 - p_2$: $-0.057 \pm 1.96 \cdot 0.0327 = (-0.121;\ 0.007)$.

5 The χ^2 test of independence

For two categorical attributes we build an $r \times c$ contingency table. Under H_0 (the attributes are independent) the expected count of a cell is $E_{ij} = \dfrac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{N}$.

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \qquad \nu = (r-1)(c-1). \tag{5.1}$$

A large value means that the observed counts are far from the expected ones. The test is always right-sided. Condition: all $E_{ij} \geq 5$.

EXAMPLE 4. SERVICE CHANNEL AND SATISFACTION

500 customers:

	Satisfied	Not satisfied	Total
Website	90	60	150
App	150	50	200
Branch	85	65	150
Total	325	175	500

Expected: website $97.5/52.5$; app $130/70$; branch $97.5/52.5$ (for example, $E_{11} = 150 \cdot 325/500 = 97.5$). The terms $\frac{(O-E)^2}{E}$: $0.58,\ 1.07;\ 3.08,\ 5.71;\ 1.60,\ 2.98$. $\chi^2 = 15.02$, $\nu = 2$, critical value 5.991, $p = 0.0005$. We reject independence: the share of satisfied customers differs by channel (60%, 75%, 56.7%). The cell contributions show that the main difference is the app customers. The strength of association by Cramér's $V = \sqrt{\chi^2/(N \min(r-1, c-1))} = 0.17$ — weak.

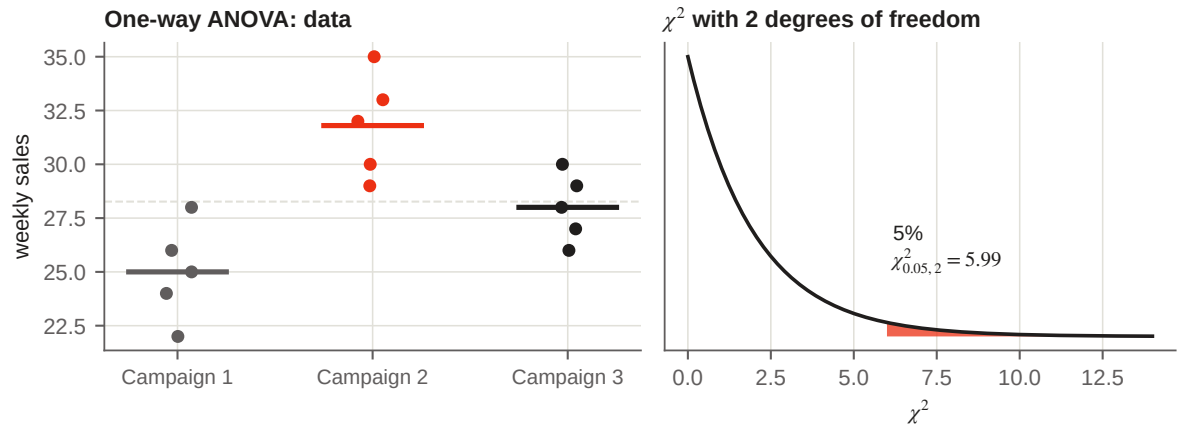


Fig. 1. Left: data of three advertising campaigns (dots) and their means; the dashed line is the grand mean. Right: the χ^2 distribution with 2 degrees of freedom and the rejection region at $\alpha = 0.05$.

5.1 The χ^2 goodness-of-fit test

It tests whether the distribution of a categorical variable matches a given one: $\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$, $\nu = k - 1$ (minus the number of estimated parameters).

EXAMPLE 5. WORKLOAD BY DAY

A manager assumes that requests are uniformly distributed over the five working days. In a week there are 120 requests: $15, 20, 24, 22, 39$. Expected $120/5 = 24$ per day. The terms: $3.375;\ 0.667;\ 0;\ 0.167;\ 9.375$; $\chi^2 = 13.58$, $\nu = 4$, critical value 9.488, $p = 0.0088$. The hypothesis of uniformity is rejected — the main contribution comes from Friday.

6 Comparing several means: ANOVA

For three or more groups, multiple t-tests lead to an inflated Type I error (Lecture 10). Analysis of variance tests $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ with one test. The idea: compare the variability *between* group means with the variability *within* groups.

$$SSB = \sum n_j(\bar{x}_j - \bar{x})^2, \quad SSW = \sum\sum (x_{ij} - \bar{x}_j)^2, \quad SST = SSB + SSW, \tag{6.1}$$

$$F = \frac{MSB}{MSW} = \frac{SSB/(k-1)}{SSW/(N-k)}, \qquad \nu_1 = k - 1,\ \nu_2 = N - k. \tag{6.2}$$

If H_0 is true, both variance estimates are close and $F \approx 1$; a large F means that the group means differ more than chance allows. Conditions: independence, approximate normality within groups, similar variances (the ratio of the largest s to the smallest no more than 2).

EXAMPLE 6. THREE ADVERTISING CAMPAIGNS

Sales over five weeks: C1: $22, 25, 24, 28, 26$ ($\bar{x} = 25.0$); C2: $30, 33, 29, 35, 32$ (31.8); C3: $27, 26, 30, 29, 28$ (28.0). Grand mean 28.27.

Source	SS	ν	MS	F
Between groups	116.13	2	58.07	**13.20**
Within groups	52.80	12	4.40	
Total	168.93	14		

Critical value $F_{0.05;2,12} = 3.885$; $p = 0.0009$. We reject H_0: mean sales differ. The share of explained variability is $\eta^2 = SSB/SST = 0.69$.

ANOVA says only that at least one pair of means differs. To find out which one, pairwise comparisons with a multiplicity correction (Bonferroni, Tukey) are used. In the example with $MSW = 4.4$ ($\nu = 12$), pairwise t-tests with the Bonferroni correction (multiplying p by 3): C1 vs C2: $p = 0.0008$; C2 vs C3: $p = 0.043$; C1 vs C3: $p = 0.129$ — the difference between C1 and C3 is not proven.

7 Practical remarks

1. Significance $\neq$ importance: report the difference of means and the CI.
2. Check the conditions: outliers and strong skewness with small n make t-tests unreliable; for χ^2 — expected counts ≥ 5.
3. An experiment with random assignment to groups permits causal conclusions; an observational comparison does not (Lecture 13).
4. A null result ($p > \alpha$) is "no evidence of a difference", not "evidence of no difference"; look at the width of the CI and the power.

8 Key terms

Term	*In Russian*	Meaning
Independent samples	*Независимые выборки*	different objects in the groups
Paired samples	*Парные выборки*	the same objects, two measurements
Pooled proportion	*Объединённая доля*	$\hat{p} = (x_1 + x_2)/(n_1 + n_2)$
Contingency table	*Таблица сопряжённости*	counts for two attributes
Expected count	*Ожидаемая частота*	$E = \text{row} \times \text{column}/N$
Analysis of variance (ANOVA)	*Дисперсионный анализ*	comparing $k \geq 3$ means
F-statistic	*F-статистика*	MSB/MSW

Formulas of Lecture 11

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_1^2/n_1 + s_2^2/n_2}}, \qquad t = \frac{\bar{d}}{s_d/\sqrt{n}}, \qquad z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})(1/n_1 + 1/n_2)}};$$

$$\chi^2 = \sum \frac{(O-E)^2}{E}, \quad \nu = (r-1)(c-1); \qquad F = \frac{SSB/(k-1)}{SSW/(N-k)}.$$

9 Practice problems

> Problem level is shown in square brackets: **[A]** — basic, tests understanding of definitions; **[B]** — standard computation; **[C]** — advanced, requires a derivation or an interpretation. Decimals are written with a point, as in a calculator, Excel and R. If a problem needs tables, use the appendix “Statistical Tables”.

Problem 1 [A] Choose the test (two-sample t, paired t, z for two proportions, χ^2 independence, χ^2 goodness of fit, ANOVA): (a) the average bill in Tashkent, Samarkand and Bukhara; (b) the weight of 20 customers before and after a programme; (c) the share of satisfied customers among men and among women; (d) the relationship between education level (3 categories) and the type of tariff chosen (4 types); (e) the average response time of two groups of operators; (f) whether the distribution of orders over the days of the week is uniform.

Problem 2 [A] Group 1: $n = 40$, $\bar{x} = 50$, $s = 8$; group 2: $n = 50$, $\bar{x} = 46$, $s = 10$. Test $H_0 : \mu_1 = \mu_2$ (H_1 two-sided, $\alpha = 0.05$) and construct a 95% CI for the difference.

Problem 3 [A] The balance of 6 customers (thousand UZS) before and after an advertising campaign: before $40, 52, 38, 45, 60, 48$; after $44, 55, 37, 50, 66, 49$. Test whether the balance increased (two-sided test, $\alpha = 0.05$).

Problem 4 [B] Branch A: $n = 15$, $\bar{x} = 78$, $s = 12$; branch B: $n = 18$, $\bar{x} = 70$, $s = 9$ (customer-service scores). Carry out Welch’s test for $H_0 : \mu_A = \mu_B$ and construct a 95% CI for the difference. Compare with the conclusion from the CI.

Problem 5 [B] Defect rate on line A: 18 out of 300; on line B: 33 out of 400. Test $H_0 : p_A = p_B$ and construct a 95% CI for the difference.

Problem 6 [B] Of 200 borrowers who received a loan under scheme A, 45 became overdue; of 200 under scheme B — 30. Carry out a χ^2 test of independence (scheme and overdue), $\alpha = 0.05$, and compare with the z-test for two proportions.

Problem 7 [B] A survey of 350 customers: region (North, Centre, South) and preferred product (loan, deposit, card):

	Loan	Deposit	Card
North	40	30	30
Centre	50	60	40
South	30	20	50

Test the independence of region and product ($\alpha = 0.05$). In which cell is the largest deviation?

Problem 8 [B] A die was rolled 60 times; faces $1, \dots, 6$ came up $8, 12, 9, 11, 6, 14$ times. Test the fairness of the die ($\alpha = 0.05$).

Problem 9 [B] One-way ANOVA: 3 groups of $n = 6$ ($N = 18$); means $20, 24, 28$; $MSW = 16$.

(a) Find SSB, MSB and F.
(b) Find the critical value $F_{0.05}$ and draw a conclusion.

Problem 10 [B] The time to perform an operation in three crews (4 observations each): A: $10, 12, 11, 13$; B: $14, 15, 13, 16$; C: $12, 11, 13, 12$. Build the ANOVA table and draw a conclusion ($\alpha = 0.05$).

Problem 11 [C] For two groups ANOVA is equivalent to a two-sample t-test with a pooled variance: $F = t^2$. Verify on the data of crews A and B from the previous problem.

Problem 12 [C] In the staff-training data (Lecture 11) $s_{\text{before}} = 6.34$, $s_{\text{after}} = 6.59$, and the correlation between scores before and after is $r = 0.975$. Explain, via the variance of the difference, why the paired test is much more powerful than the independent one. Find the SE for both approaches.